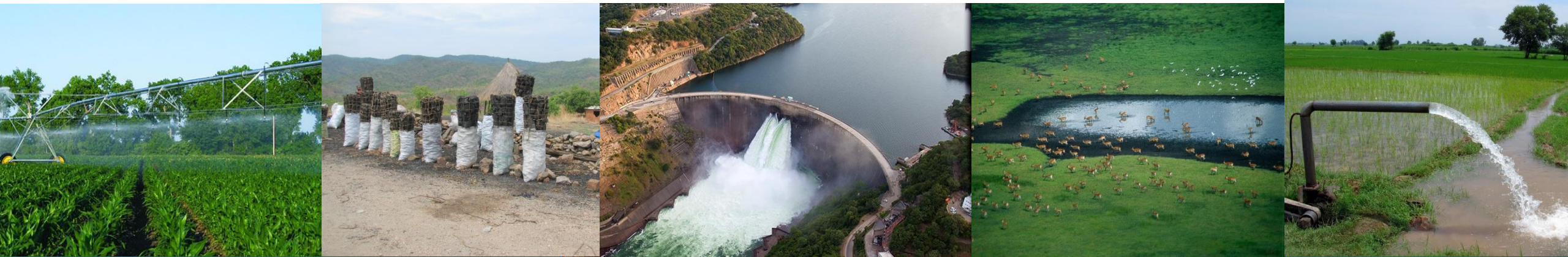


Exercise 3b: Calibration

Peter Burek, Carla Catania, Silvia Artuso
International Institute for Applied Systems Analysis
Water Security Group



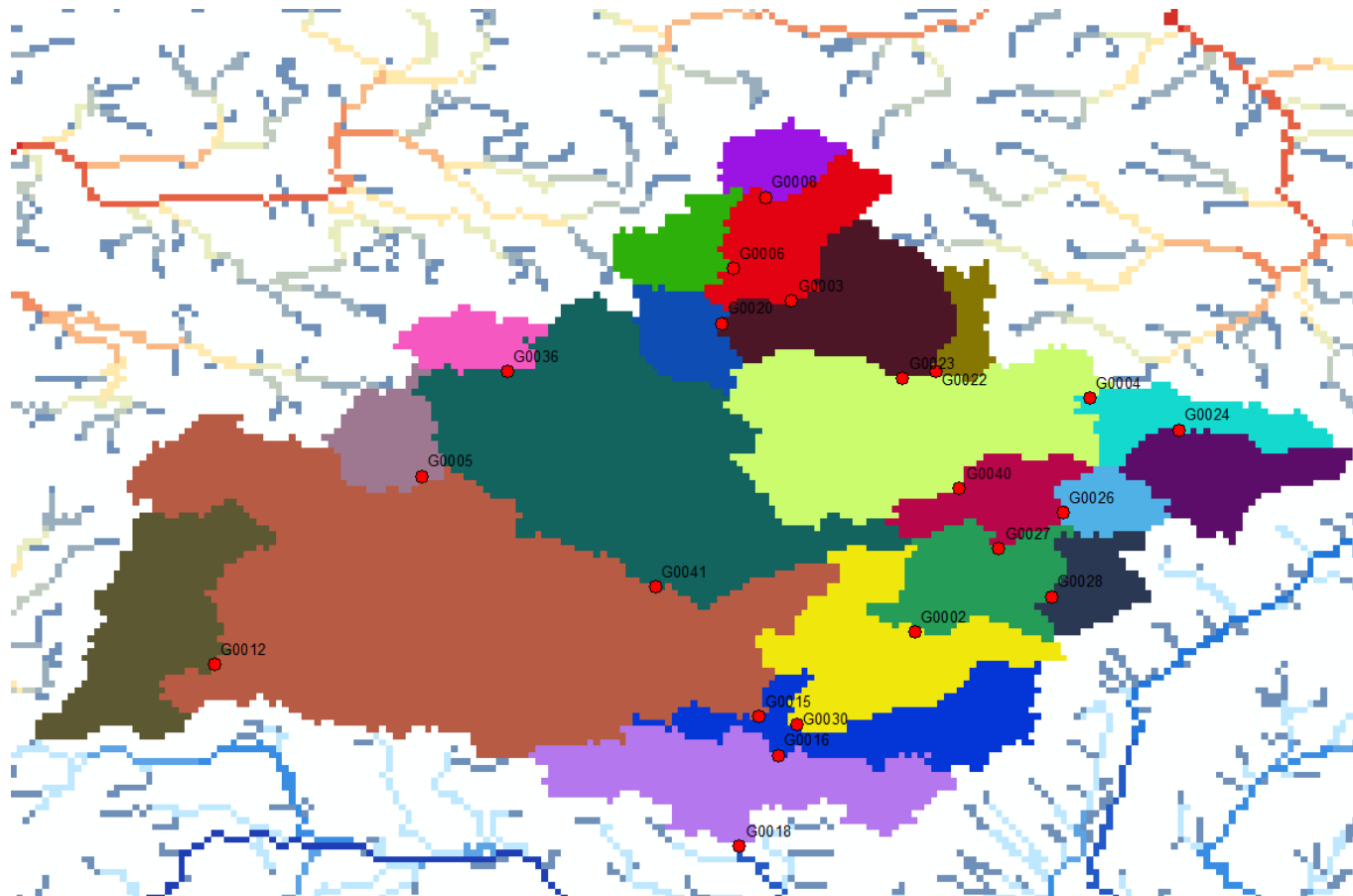
Calibration

1. Upstream to downstream calibration
2. Preparing settings files
3. Running a calibration
4. Visualization of the results
5. Creating parameter maps



Calibration

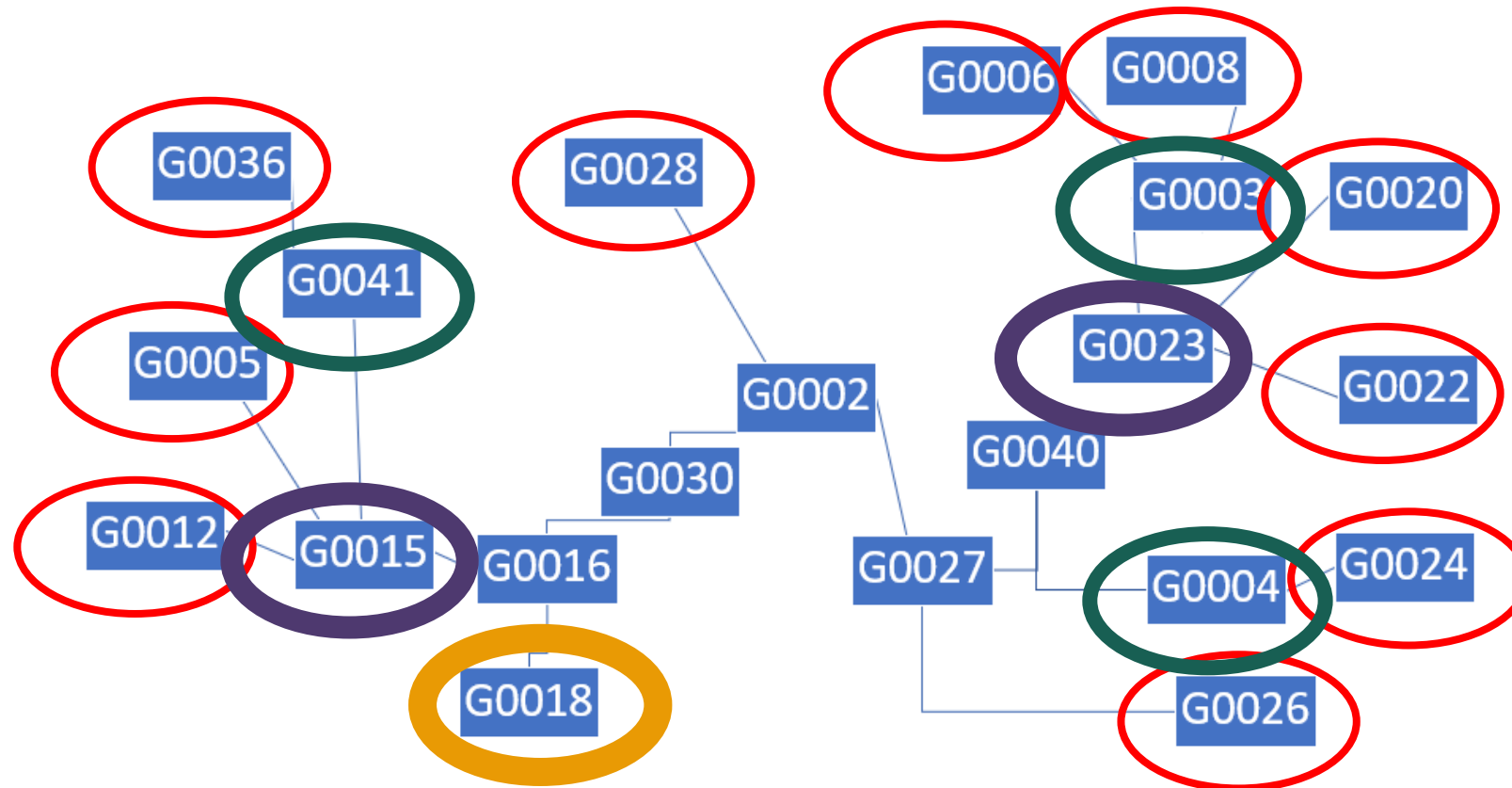
Upstream to downstream calibration



Morava basin with 21 subbasins

Calibration

Upstream to downstream calibration



Morava basin with 21 subbasins

Calibration

1. Running a fast calibration for an upstream basin

Go to your basin folder and /basin_calibration_tool

Correcting path in the settings files:

1) Settings1.txt

Set the path name to your computer settings

[DEFAULT]

```
Root = C:/work/CWATM/Morava
RootLinux = /p/luc/watmodel/calibration/danube_1min
Rootbasin = morava_calibration_tool

# start date of simulation outputs
ForcingStart = 1/1/1990
.....
```

```
44 No_of_calibration_lists = 1
45 RunCwatm = python C:/work/CWATM/Morava/morava_calibration_tool/run_cwatm.py
46 RunCalib = python C:/work/CWATM/Morava/morava_calibration_tool/calibration_sing
47 listPC = listPC
48
```

Set this to: use_multiprocessing = 1

```
60 # if more than one node is to be used for parallel processing
61 use_multiprocessing = 1
62
```

Calibration

2. Running a calibration

Go to your basin folder and `/basin_calibration_tool/`

Running a fast calibration

Execute: `runmain.bat` or `python calibration_main2.py`

This will launch a (fast) upstream to downstream calibration

(Fast because the [DEAP] parameter are set to only a few runs)

Calibration

2. Running a calibration

The fast calibration is using only:

- Initial population of 4 (normally ≥ 256)
- 2 generations (normally ≥ 10)
- 2 runs per generation (normally ≥ 32)

- ➔ the calibration is splitting the runs for multiprocessing
- ➔ Your computer is storing the progress of the subbasin in the folder: listPC
- ➔ You can use several computers to run the same basin. Each computer will pick a new subbasin which is
 - not already run
 - which is not run by another computer
 - which is next in the line of upstream-downstream
 - after some time, it will redo unfinished subbasin

Settings1.txt – last part

```
[DEAP]
maximize = True

# if more than one node is to be used for parallel processing
use_multiprocessing = 0

# Number of nodes used for multiprocessing, recommended 1-8
pool_limit = 32

# Number of generations, recommended 10-30
ngen = 10
ngen = 2

# Initial population, recommended 128-256
#mu = 256
mu = 4

# Runs per generation, recommended 8-32
#lambda_ = 32
lambda_ = 2

# Select the best out of the generation
#select_best = 10
select_best = 2
```

Calibration

2. Running a calibration

The fast calibration is using only:

- Initial population of 4 (normally ≥ 256)
- 2 generations (normally ≥ 10)
- 2 runs per generation (normally ≥ 32)

Please stop the run

Change back:

Use_multiprocessing = 1

➔ the calibration is splitting the runs for multiprocessing

Run again: [runsingl.e.bat](#)

Folder structure in:

morava_calibration_tool\calibration1\G0022\out
at start:

Name ^

00_000
00_002
00_003
00_004

Calibration

3. Look at the results

Your DOS box should look like this

```
run_rand_id: 00_002, KGE: 0.469, COR 0.571, NSE: 0.203
run_rand_id: 00_004, KGE: 0.529, COR 0.571, NSE: 0.210
>> gen: 0, KGE: 0.529
run_rand_id: 01_001, KGE: 0.552, COR 0.578, NSE: 0.177
run_rand_id: 01_002, KGE: 0.509, COR 0.554, NSE: 0.249
>> gen: 1, KGE: 0.552
run_rand_id: 02_002, KGE: 0.469, COR 0.571, NSE: 0.260
run_rand_id: 02_001, KGE: 0.553, COR 0.585, NSE: 0.204
>> gen: 2, KGE: 0.553
>> Termination criterion ngen fulfilled.
>> Saving optimization history (front_history.csv)
>> Saving Pareto optimal solutions (pareto_front.csv)
>> Running Model using the "best" parameter set
CWATM - Community Water Model 1.5 Date: 01.06.2021
International Institute of Applied Systems Analysis (IIASA)
Running under platform: Windows
-----

CWATM Simulation Information and Setting
The simulation output as specified in the settings file: P:\watmodel\CWATM\Regions\Danube_1min_SP\Morava
and in P:\watmodel\CWATM\Regions\Danube_1min_SP\Morava\morava_calibration_tool\calibration1\G0022\out\03
rava_1min2-Run03_best.ini

Step      Date      Discharge
12053  31/12/2022      1.73
Start: 10:42:24.688515
End: 10:50:39.475264
>> Saving "best" simulated streamflow (streamflow_simulated_best.tss)
=====DONE == G0022 =====
```

Folder structure in:

morava_calibration_tool\calibration1\G0022\out
at end of the calibration:

-  inflow
-  maps
-  out
-  checkpoint.pkl
-  front_history.csv
-  pareto_front.csv
-  runs_log.csv
-  streamflow_simulated_best.tss

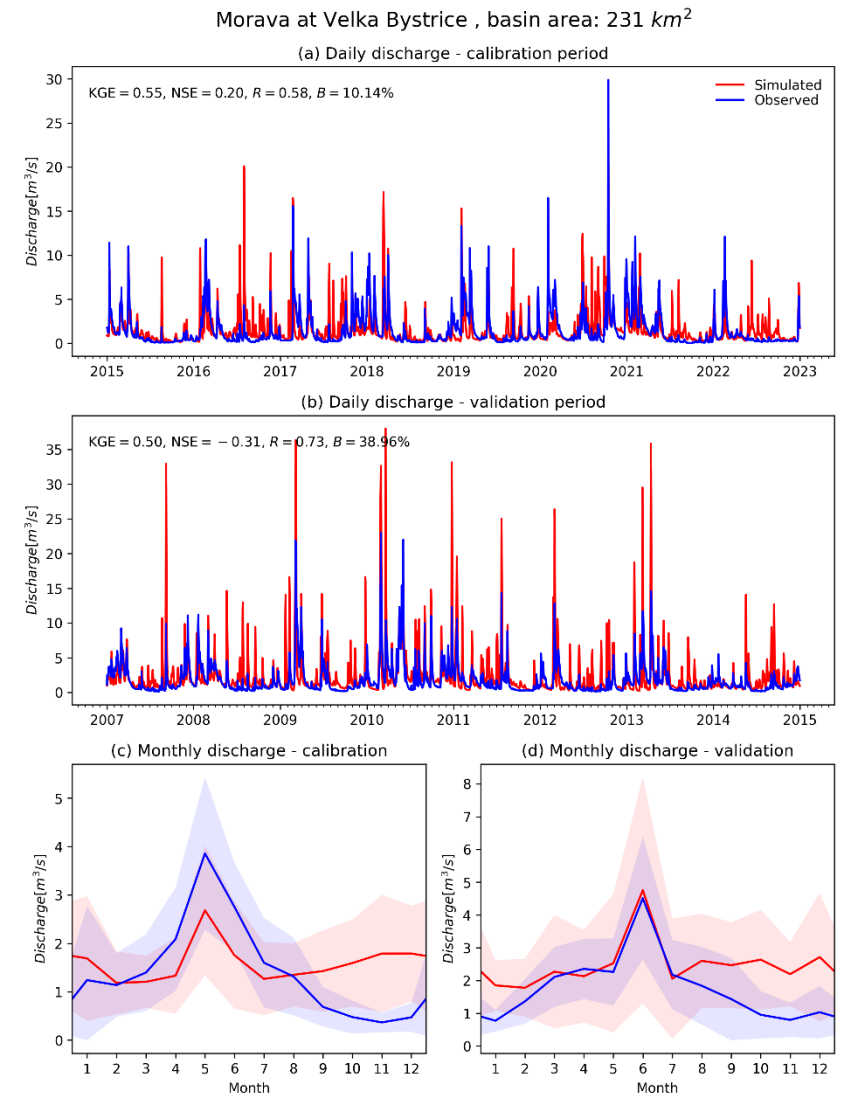
After each subbasin calibration, the subbasin is run again for a greater time period, in case that the downstream basin uses another calibration period

Calibration

3. Look at the results

Execute: `runC10.bat` or
`python CAL_10_COMPUTE_STATS_AND_FIGURES_TIME_SERIES.py settings1_fast.txt`

To create a figures in:
`yourbasin_calibration_tool\calibration1\FIGURES`



G0022_Morava_Velka_Bystrice_.png

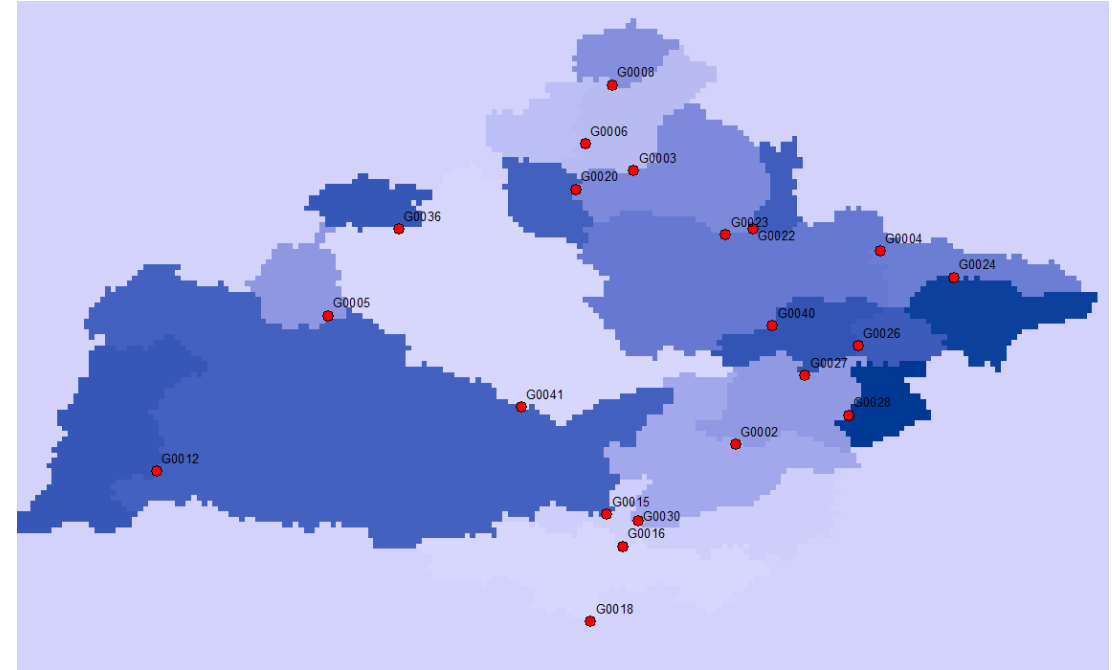
Calibration

4. Create parameter maps

`python CAL_9_PARAMETER_MAPS.py`

This create parameter maps in: results
e.g. params_arnoB.map

- you have to change the path to the settings file inside CAL_9_PARAMETER_MAPS.py
- you have to install the PCRaster Python library (we working to get rid of this library, which will make this much easier)



Map of arnoB parameter for Morava

Calibration

4. Putting parameter maps in CWatM

Parameter can be a number:

SnowFactor = 1.2

But also a spatial map.

The parameter maps can be read
as spatial maps into CWatM

In the section [Calibration]

SnowFactor = \$(FILE_PATHS:PathParameter)/params_snowfactor.map

[CALIBRATION]

```
# These are parameter which are used for calibration
# could be any parameter, but for an easier overview, tehey are collected here
# in the calibration template a placeholder (e.g. eta) instead of value

# reservoir normal storage limit (fraction of total storage, [-]) [0.15 - 0.85] default 0.5
# These are parameter which are used for calibration
# could be any parameter, but for an easier overview, tehey are collected here
# in the calibration template a placeholder (e.g. 0.11805799193919307eta) instead of value

SnowFactor = $(FILE_PATHS:PathParameter)/params_snowfactor.map
SnowMeltCoef = $(FILE_PATHS:PathParameter)/params_snowMelt.map
#IceMeltCoef = $(FILE_PATHS:PathParameter)/params_iceMelt.map

crop_correct = $(FILE_PATHS:PathParameter)/params_crop.map
preferentialFlowConstant = $(FILE_PATHS:PathParameter)/params_pref.map
arnoBeta_add = $(FILE_PATHS:PathParameter)/params_arnoB.map
ksat_fact = $(FILE_PATHS:PathParameter)/params_ksatfact.map

factor_interflow = $(FILE_PATHS:PathParameter)/params_interF.map
recessionCoeff_factor = $(FILE_PATHS:PathParameter)/params_reces.map

manningsN = $(FILE_PATHS:PathParameter)/params_CCM.map
#manningsN = 1.0

normalStorageLimit = $(FILE_PATHS:PathParameter)/params_normalStor.map
lakeAFactor = $(FILE_PATHS:PathParameter)/params_lakeAFactor.map
soildepth_factor = $(FILE_PATHS:PathParameter)/params_soilfactor.map

# runoff concentration factor runoffConc_factor = 1.0
#runoffConc_factor = 1.0
runoffConc_factor = $(FILE_PATHS:PathParameter)/params_runoff.map

IceMeltCoef = 0.005
# lake parameter - factor for wind evaporation
lakeEvaFactor = 1.2
```

Calibration

5. Next steps (not this time)

Preprocessing:

- putting metadata of stations and timeseries of discharge in /observed_data
- calculating which subbasins is upstream of another
- calculating the subbasins (running a basin – upstream basin + inflow from subbasins)

Calibration

What can go wrong 1

Run runsingle.bat

`python calibration_single_single.py`

- Your python version is starting with python3 or you have to put in the full path of your python version
c:/python38/python3 calibration_single_single.py
- You did not change the path in settings file settings1_fast.txt and templates/settings_morava_1min2.ini
- You did not download the meteo data from Sharepoint or you did not download other data
- You still have problems with the gdal library
 - use the newest version of CWatM on Sharepoint
 - install gdal like in <https://cwatm.iiasa.ac.at/setup.html#external-libraries>
- You have problems with white space in file names
Use e.g. "c:/white space/python.exe" calibration_single_single.py



Calibration

What can go wrong 2

Run runsingle.bat

`python calibration_single_single.py`

- You did not add the libraries deap, matplotlib and pandas
- Some results are already in basin/calibration_tool/calibration1/G0022 (or the G00?? You are using)
- delete the files and the /out directory, make a new out directory



Calibration

1. What can go wrong 3

Run runsingle.bat

`python calibration_single_single.py`

You run it but it crash:

- Go to basin_calibration_tool/calibration1/G00??/out
- Is there a folder 00_000?
 - No: bad, go back to what can go wrong 2
 - yes: good, go into 00_000
- Inside 00_000 there is a file runpy00_000.bat
 - type cmd in the explorer to start a new DOS box
 - type runpy00_000.bat and execute

CWatm is running?

Yes: very good – check if the observed data in

basin_calibration_tool \observed_data/ qgis_yourbasin_5.csv are in right format

No: change settings_fast-Run00_000.ini (e.g. path name) till CWatM is running

copy the changes into templates/settings.ini

Please have a look at:

<https://cwatm.iiasa.ac.at/calibration.html>

https://cwatm.iiasa.ac.at/calibration_tutorial.html